Basic Mathematical Definitions
and
Using Python for Cryptography

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Rule of Exponentiation

$$g^{a^b} = g^{b^a} = g^{ab}$$

Exponentiation


```python
def exp():
    g = 9 # generator
    a_secret = 13
    a_public = pow(g, a_secret) # g^a_secret
    b_secret = 42
    b_public = pow(g, b_secret) # g^b_secret
    ss_alice = pow(b_public, a_secret)
    ss_bob = pow(a_public, b_secret)
    assert ss_alice == ss_bob
```

$$g^{a^b} = g^{b^a} = g^{ab}$$

Modular Exponentiation


```
def mod_exp():
    p = 3329 # generator
    g = 9
    a_secret = 13
    a_public = pow(g, a_secret, p) # g^a_secret mod p
    b_secret = 42
    b_public = pow(g, b_secret, p) # g^b_secret mod p
    ss_alice = pow(b_public, a_secret, p)
    ss_bob = pow(a_public, b_secret, p)
    assert ss_alice == ss_bob
```

$$g^{a^b} \equiv g^{b^a} \equiv g^{ab} \pmod{p}$$

Given

g, p, and $g^a \pmod{p}$,

find a

Definitions - Function and its friends

Function

Domain

Codomain

Range

Function is a mapping from a *domain* to a *codomain*.

Range is the subset of the codomain that is *actually* mapped to by the function.

Definitions - Image and Preimage

Image of a function f is the set of all outputs of f.

Preimage of a function f is the set of all inputs that map to a given output of f.

Definitions - Second Preimage

Given a function f and an input x, a **second preimage** of x is an input $x' \neq x$ such that $f(x) = f(x')$.

Definitions - One Way Function

A **one-way function** is a function that is easy to compute but hard to invert.

More formally, a function f is a one-way function if:

- f can be computed in polynomial time.
- For every probabilistic polynomial-time algorithm A, the probability that $A(f(x))$ outputs a preimage of $f(x)$ is negligible.

Hash Functions

A **hash function** is a function that maps data of arbitrary size to a fixed-size value.

A **cryptographic hash function** is a hash function that has the following properties:

- It is easy to compute the hash value for any given input.
- It is infeasible to generate a valid input that has a given hash value.
- It is infeasible to find two different inputs that produce the same hash value.

Random Number Generator

A **cryptographically secure pseudorandom number generator** (CSPRNG) is a pseudorandom number generator that has the following properties:

- It is computationally indistinguishable from a true random number generator.
- It is resistant to state compromise extensions, meaning that even if an attacker learns the internal state of the generator at some point, they cannot predict future outputs.